WEEK 15 · Deep Learning

Deep Learning Time Series Forecasting

Forecast the future from past patterns, then test it with financial decisions

LSTM · TCN · Attention · PatchTST · iTransformer · Chronos-2 · TimesFM-3 · TOTEM

Today's question — do deep learning and time series foundation models actually help financial forecasting?
Judge by comparing on the same future period, same baseline, and same decision rule.

Judge deep learning forecasts by the decisions they support

Three messages that run through this lecture

1

First fix the target and the input window

Pick targets tied to decisions, such as volatility and volume, and input only what was known at the forecast origin.

2

Different architectures, same question

LSTM, TCN, Attention, patches and foundation models differ only in how they bundle past information; the evaluation standard is the same.

3

A good forecast score is not a good financial result

Compare with simple baselines using time-ordered splits, interval coverage and trading costs.

All figures and numbers are illustrative or hypothetical — not measured performance

01

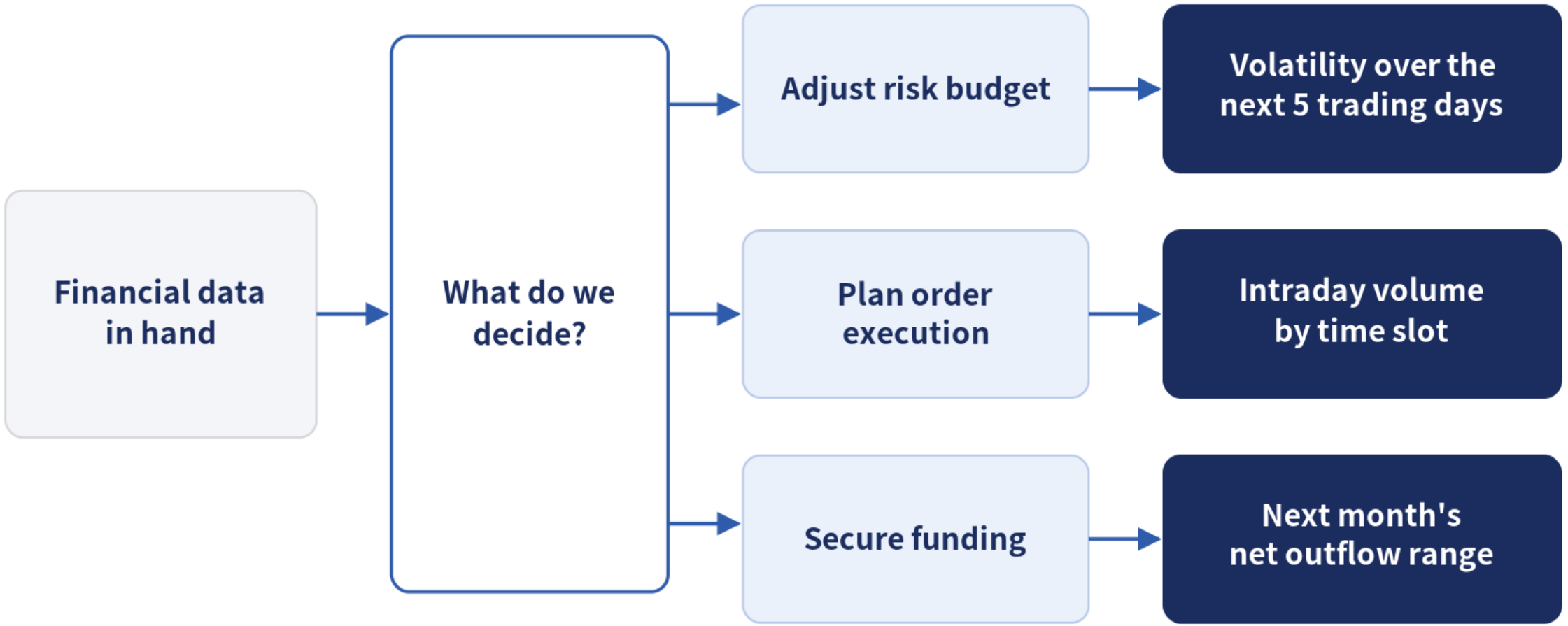
PART 1 · Basic Structure

Basic Structure of Deep Learning Forecasting

Target · input window and label · LSTM·TCN · Attention · forecast range

The target starts from the decision to be made

Start with volatility, volume and cash flows, which link to decisions more easily than price direction



Five terms define a forecasting problem

Before looking at the model, fix what goes in and what is predicted

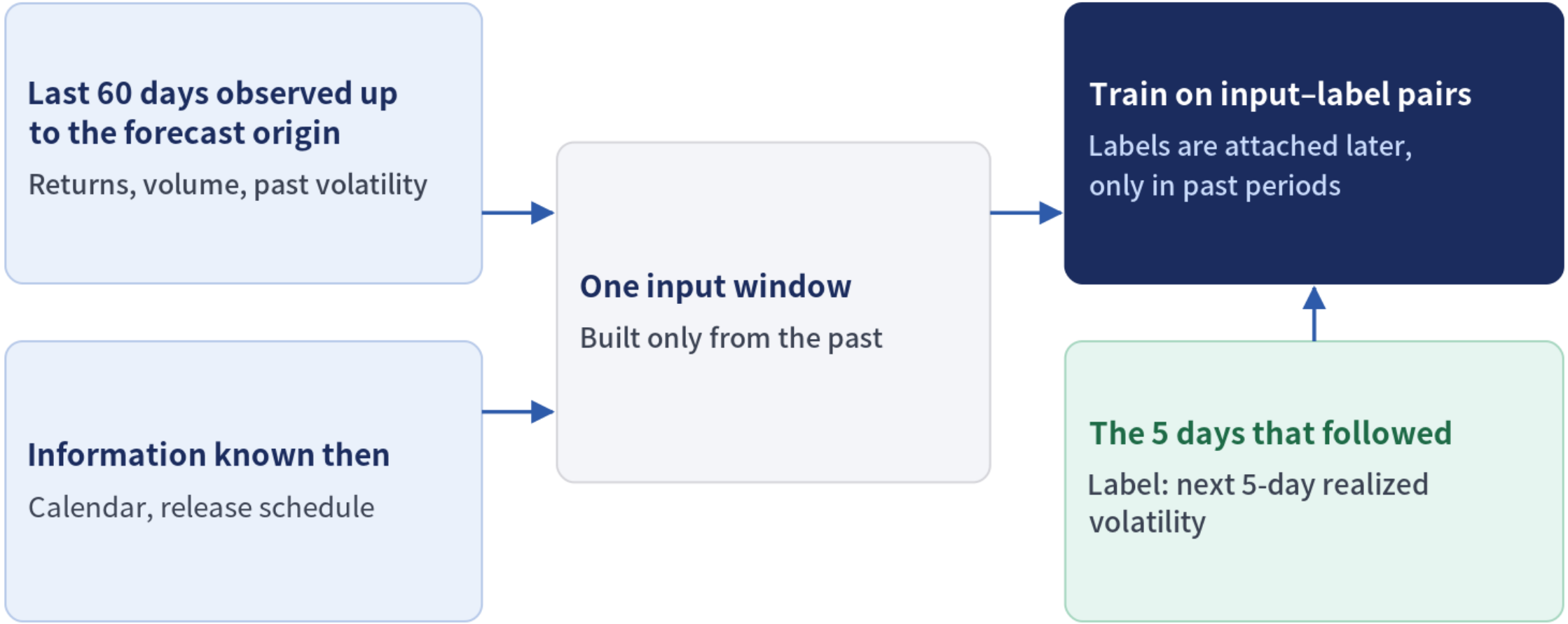
Term	Meaning	Financial example
Input window (lookback)	Past data up to the forecast origin	Last 60 days' returns, volume
Forecast horizon	Period to be predicted	Next 5 trading days
Label (target)	Value known after the horizon ends	Next 5-day realized volatility
Point · quantile forecast	One center · lower-to-upper range	Median 18% · range 12–28%
zero-shot	Applied with no task-specific training	Foundation model used as is

Rule

Labels are attached later, only in past periods where training is done — live inputs cannot contain future observations

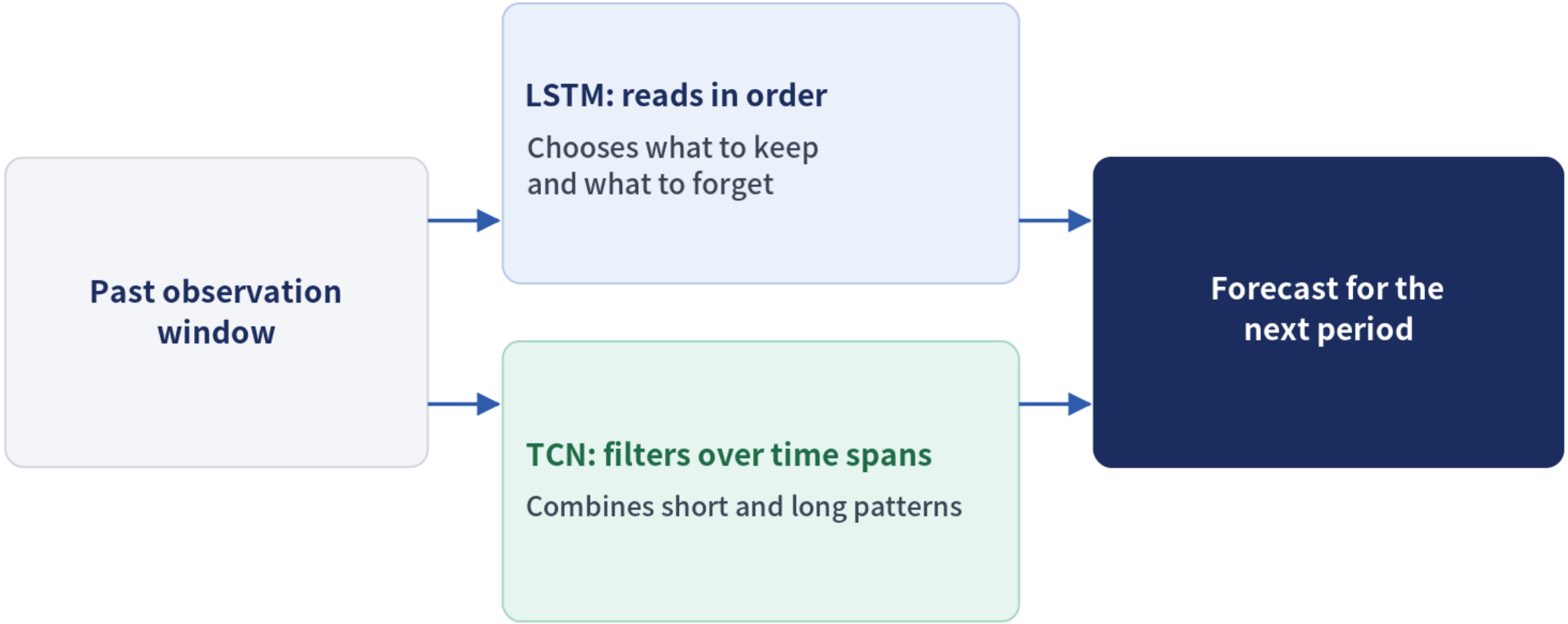
Build training data by pairing input windows with labels

Today's live input cannot contain observations from the next 5 days



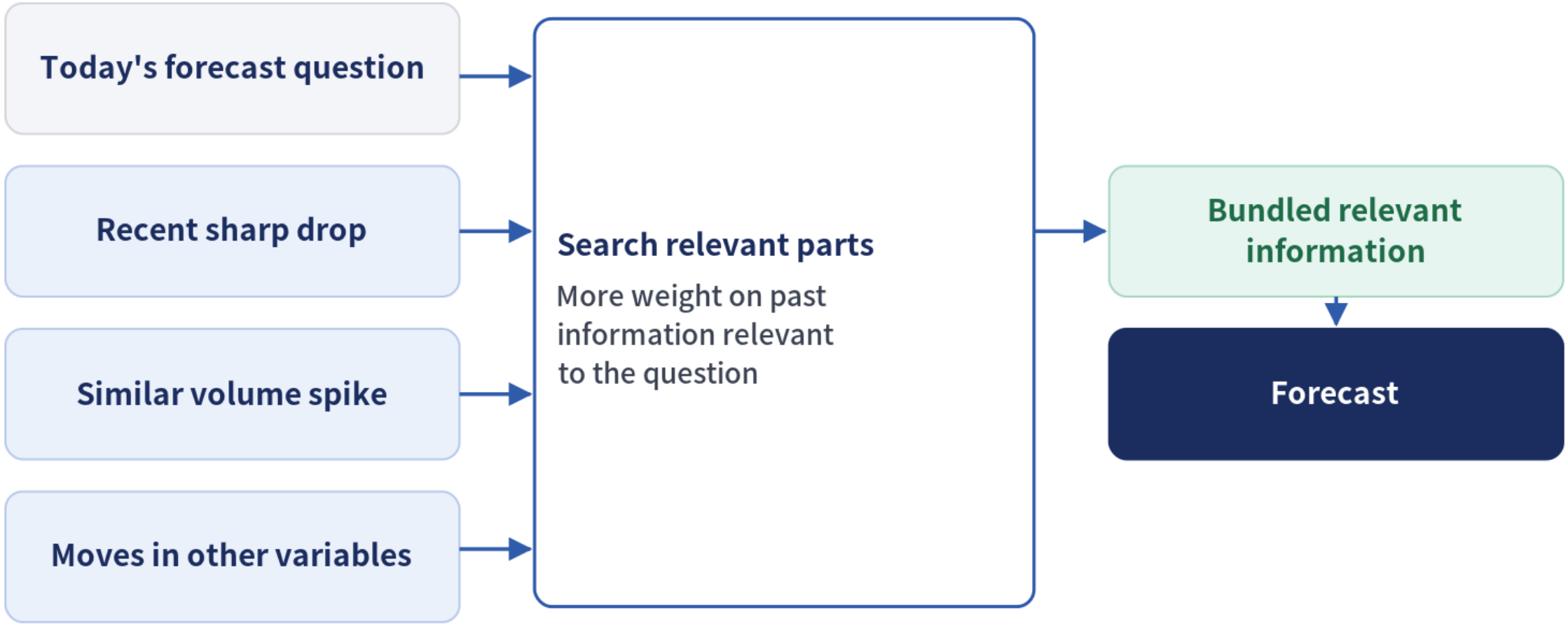
LSTM selects what to remember; TCN finds patterns

Both take a past observation window and forecast the next period



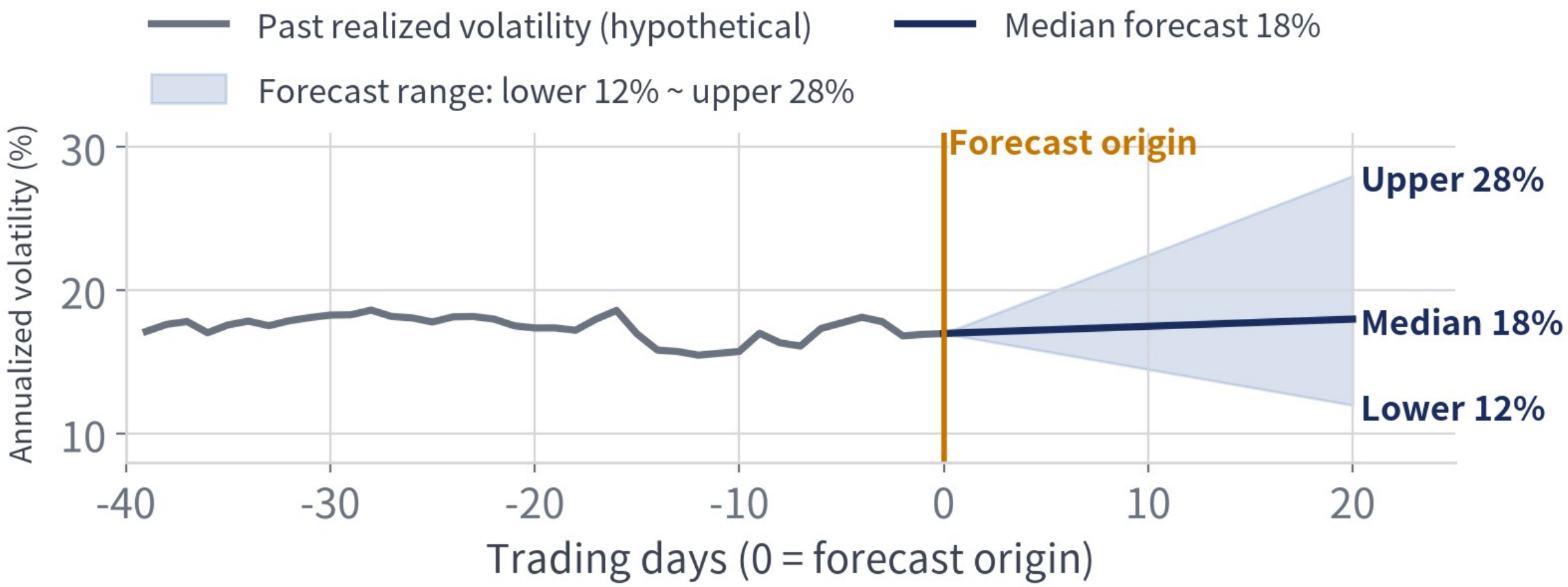
Attention selects and bundles relevant information

It gives more weight to past periods and variables relevant to the forecast question



A point forecast and a forecast range are different information

Hypothetical example: median 18%, lower 12%, upper 28% from the same past data · ※ illustrative values (annualized volatility)



Read architecture diagrams with their limits in mind

Do not read the analogies, weights or ranges in a diagram as evidence of performance or causality

Item	Common misconception	Correct check
Forecast target	Volatility and volume are easy to forecast	Compare with simple baselines first
LSTM·TCN analogy	The model remembers like an analyst	Verify what information it has learned
Attention weights	High weight = causality or alpha	Check token construction and masks
Forecast range	It is a guaranteed confidence interval	Measure coverage on the actual future

02

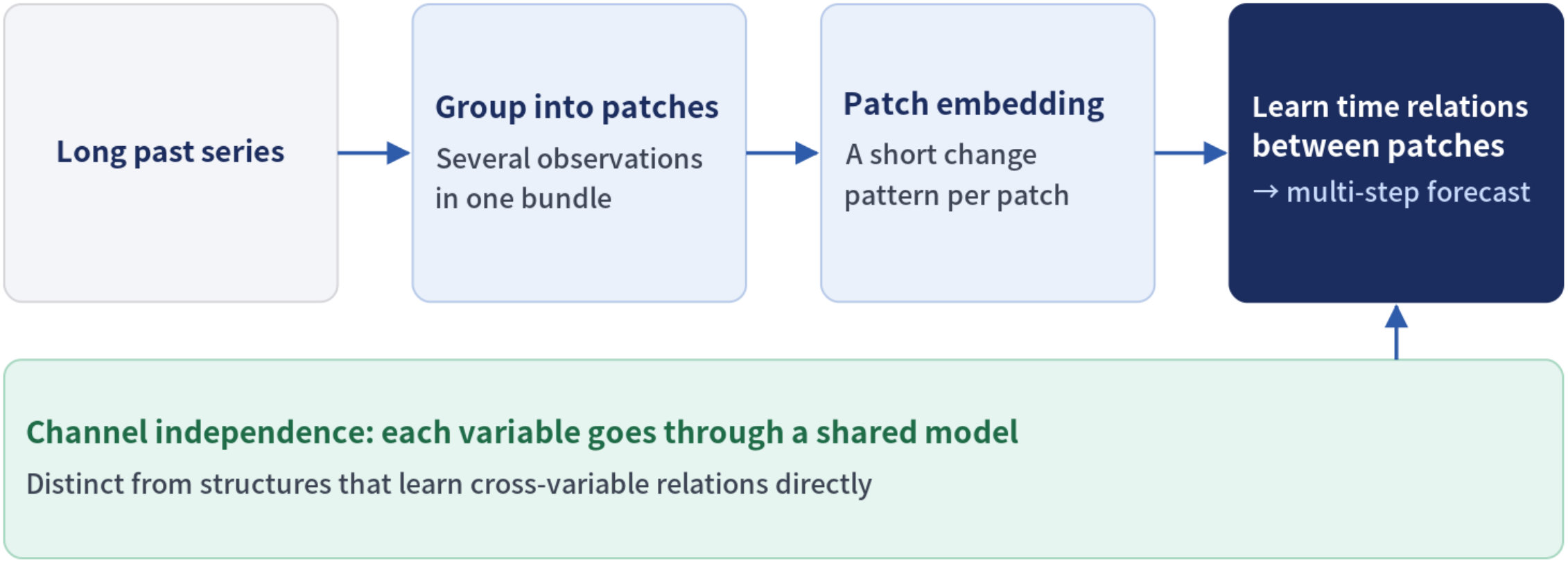
PART 2 · Recent Models

Recent Time Series Deep Learning and Foundation Models

PatchTST · iTransformer · Chronos-2 · TimesFM-3 · TOTEM

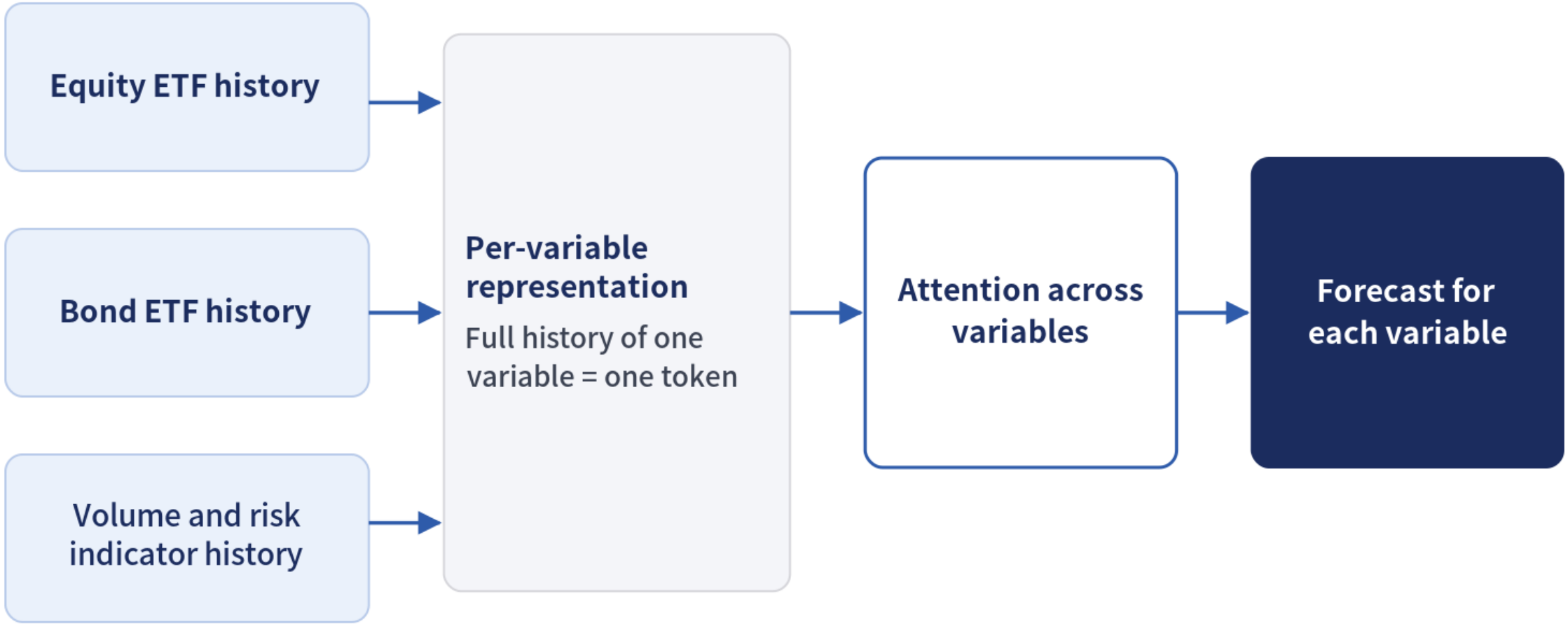
PatchTST reads a series in bundled segments

Bundles observations into patches rather than single days and learns relations between patches [S1]



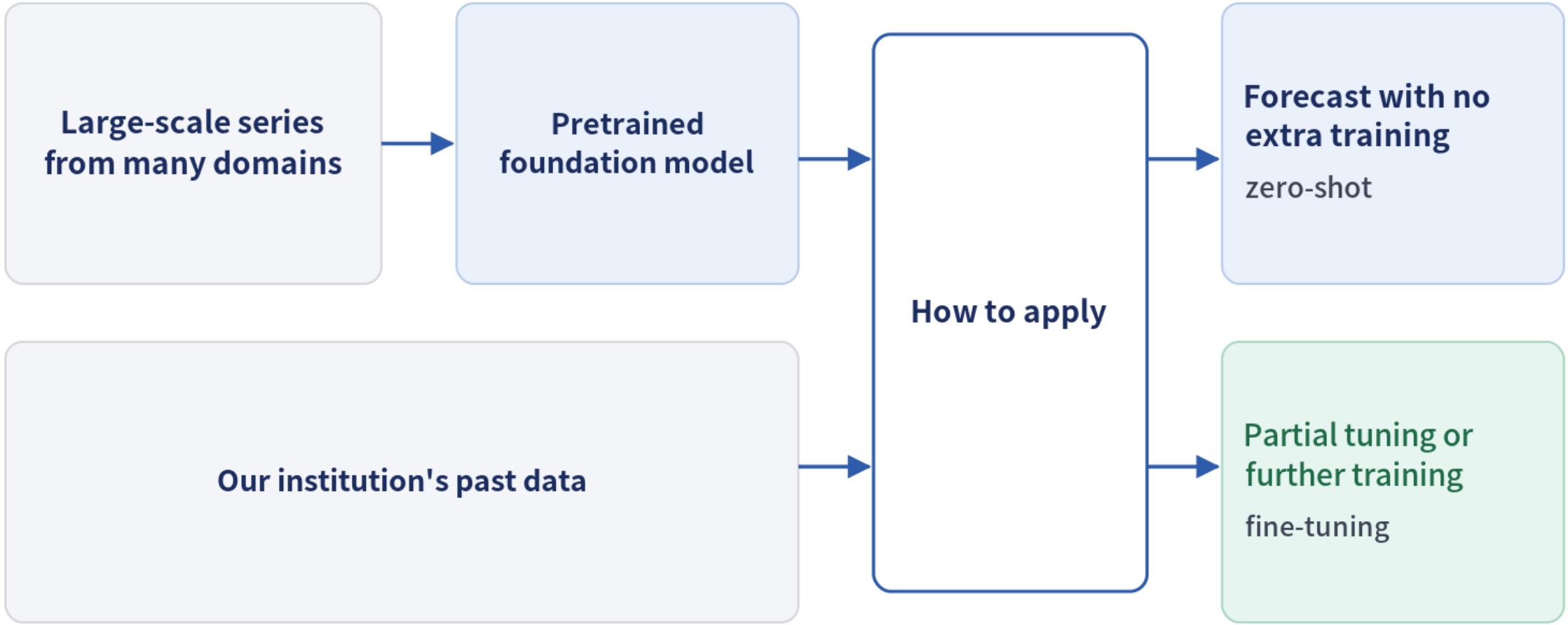
iTransformer compares each variable's whole history

A representative architecture that shifts tokens from time points to variables [S2]



Foundation models are not trained from scratch

A model pretrained on large time series from many domains is applied to in-house data as is or after tuning



Three families, each with a different role

Cases confirmed from official releases as of 2026-11-04 — not a ranking of all recent models

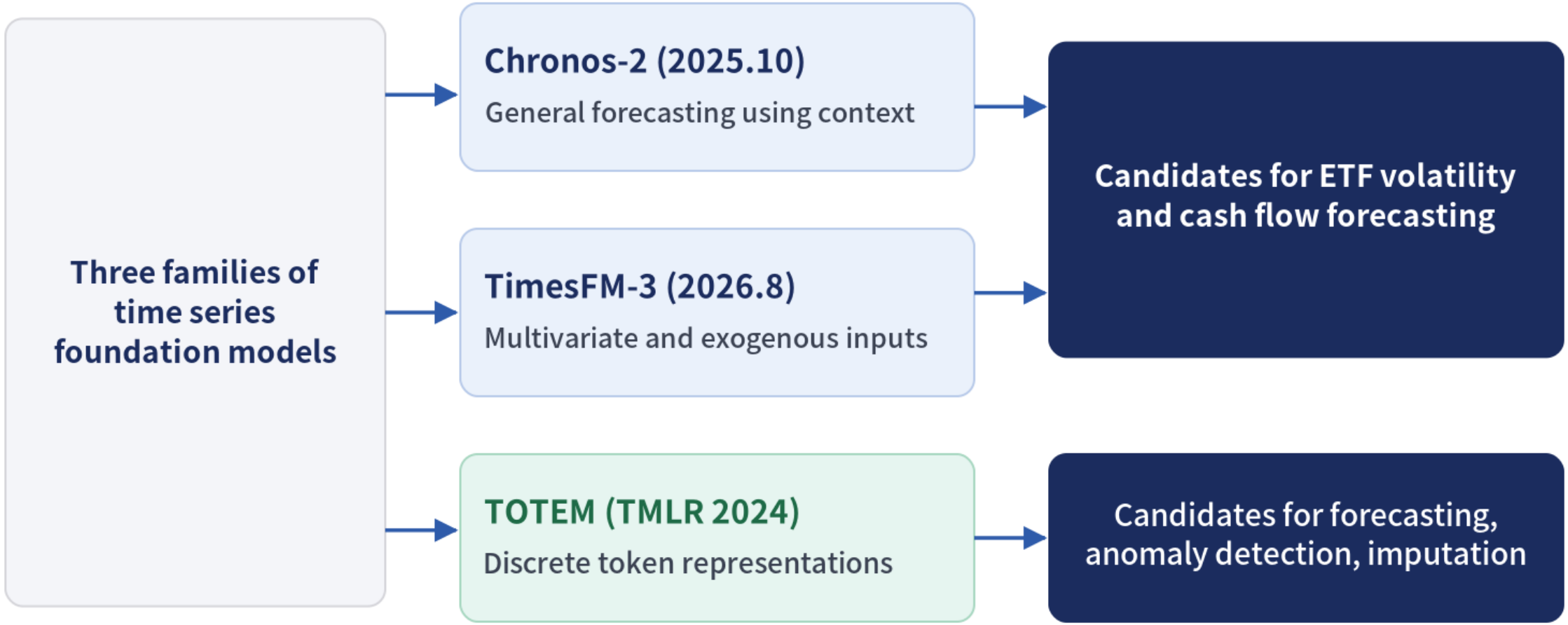
Model	Developer · venue	Key feature in the figure	Financial application
Chronos-2	Amazon · 2025	Patches of history, joined with time and group info	Range forecasts: ETF volatility, volume, cash flows
TimesFM-3	Google Research · 2026	Reads time and variable relations together	Multi-asset volatility; calendar-aware volume
TOTEM	Talukder·Yue·Gkioxari · TMLR 2024	Series as learned discrete codes	Abnormal volume, imputation, forecasting

Why these

Chronos and TimesFM are recent general-purpose forecasters; TOTEM is a representation-learning case that extends beyond forecasting

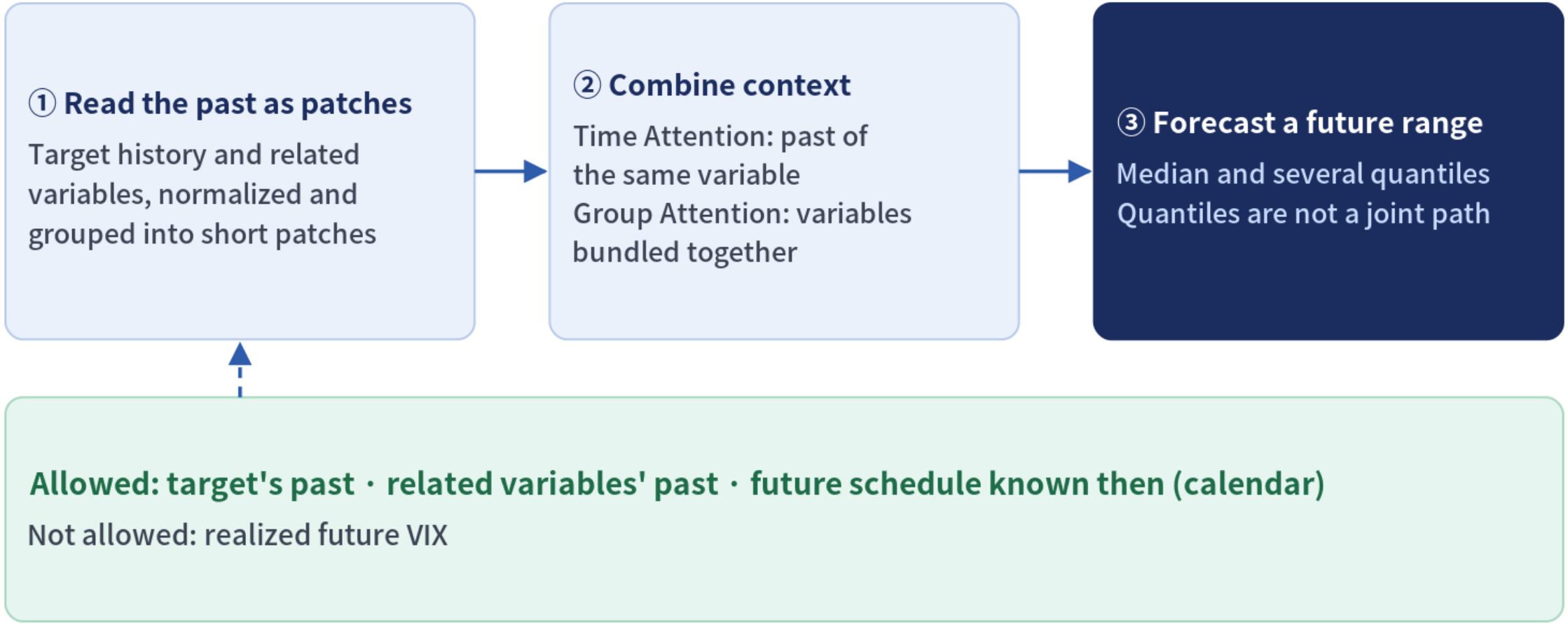
Two general forecasters and one representation learner

TimesFM-3 (2026.8) broadens multivariate and exogenous-variable support; Chronos-2 (2025.10) also uses covariates [S3][S4]



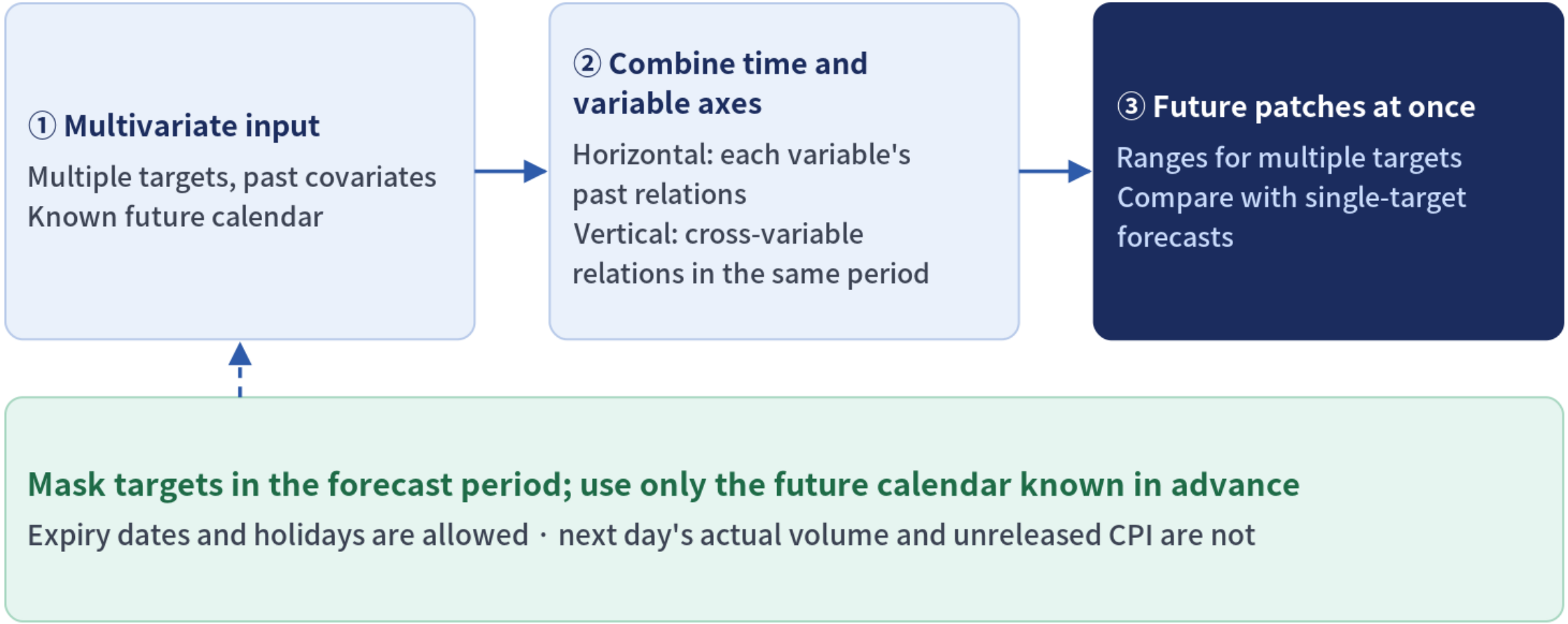
Chronos-2 forecasts ranges from history and variable context

Rescales observations, combines them with time and group Attention, and outputs multiple quantiles [S4]



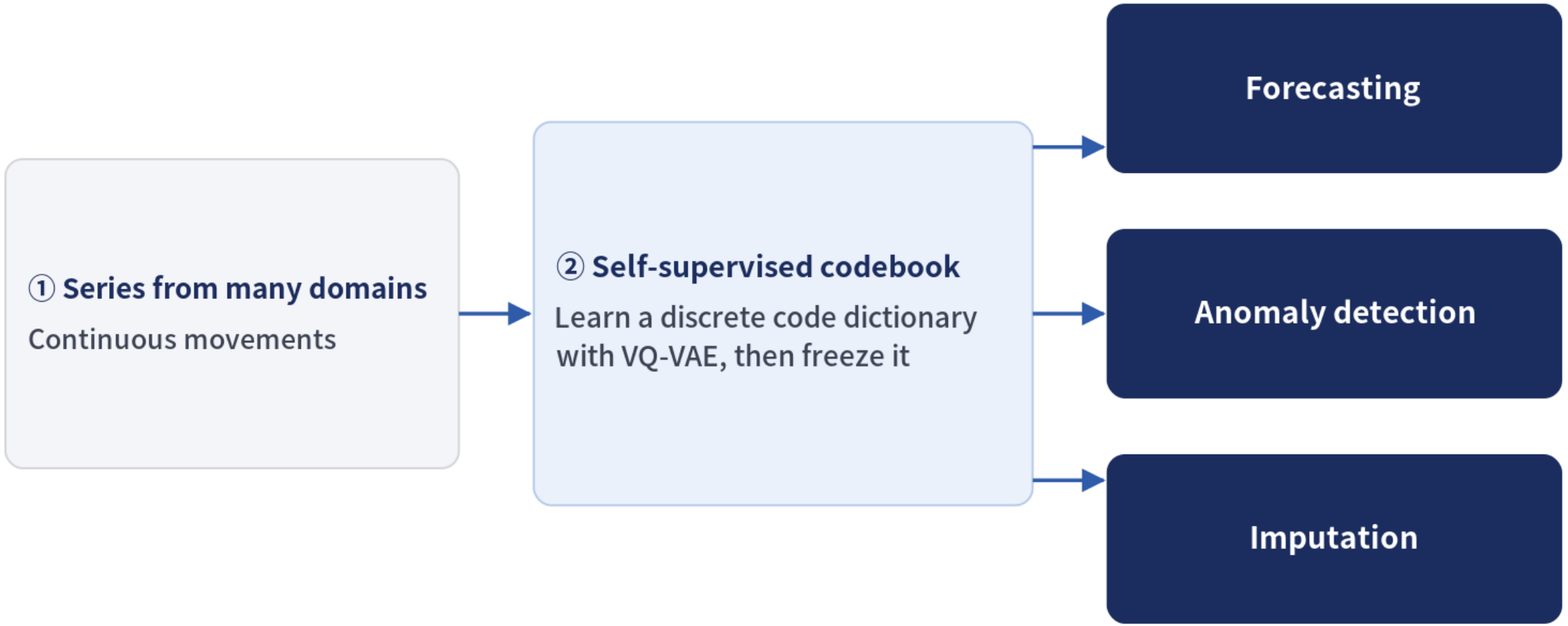
TimesFM-3 reads co-moving variables together

Patches multiple targets and past covariates, combines time and variable axes, and forecasts future patches at once [S3]



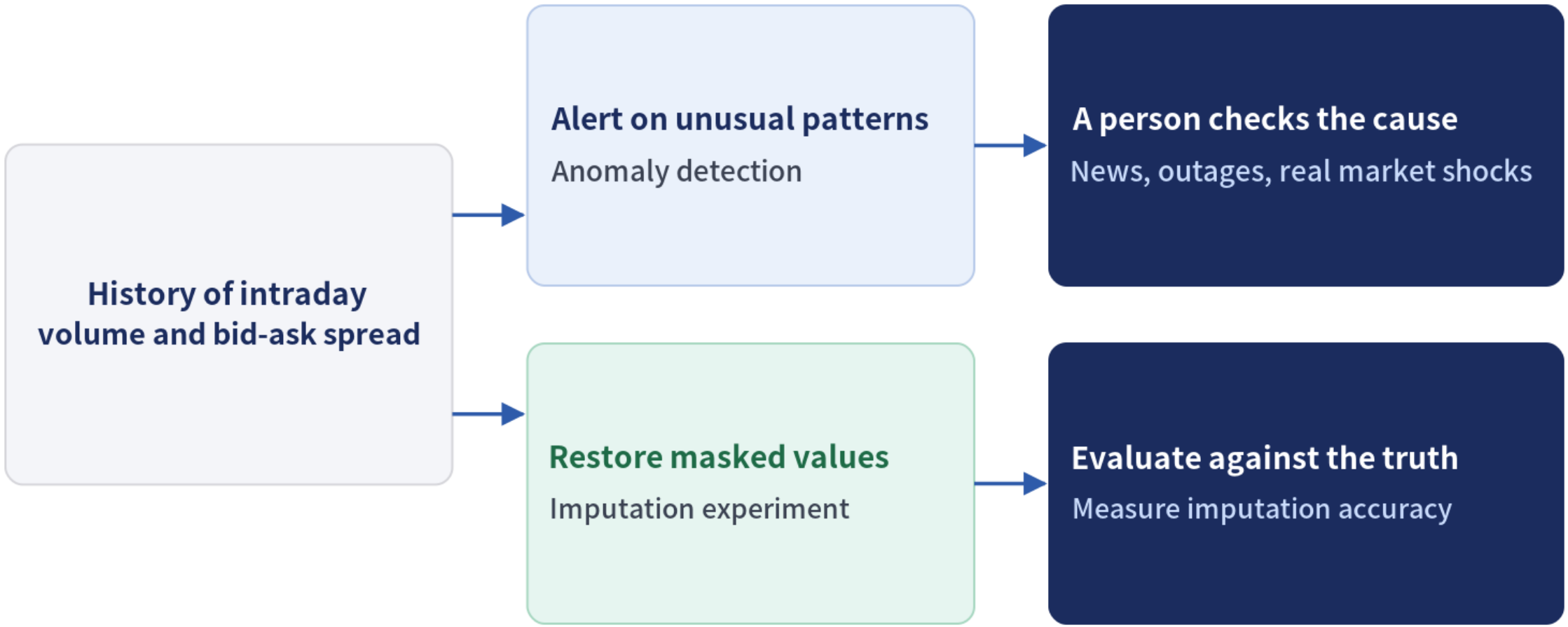
TOTEM represents movements with learned codes

Learns a codebook mapping series to discrete codes by self-supervised learning, then freezes it for many tasks [S10][S11]



TOTEM can serve for alerts and data quality checks

These are financial application designs; the original paper did not validate performance on these financial tasks



TOTEM and Toto are different models

Similar names, different families and functions — recheck version and features before running

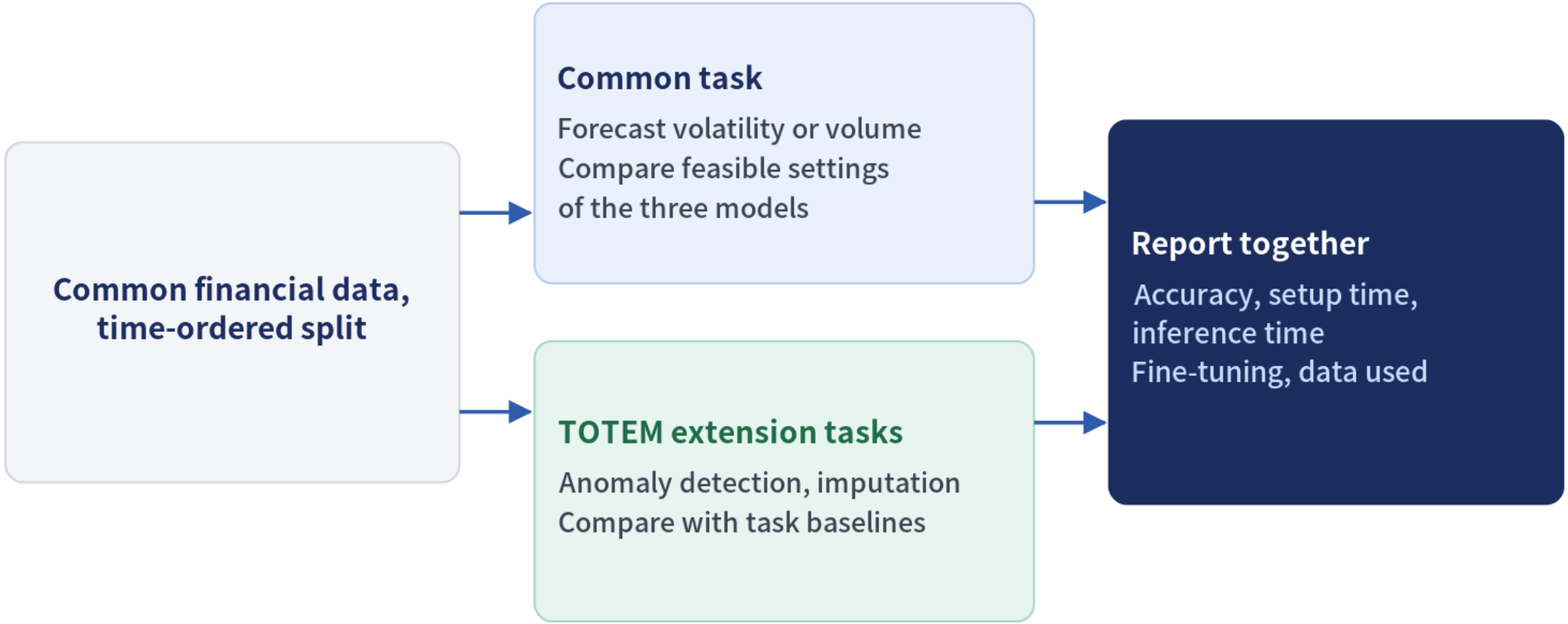
Name	What it is	Role in this lecture
TOTEM	TOkenized Time Series EMbeddings · discrete representations for many tasks	One of the three selected families
Toto	Datadog's multivariate time series forecasting foundation model	Reference for further exploration

Toto 2.0 note [S12]

- A separate family using multivariate Attention and quantile forecasts
- Per the official implementation docs, fine-tuning and exogenous-variable support for 2.0 are not yet provided — do not mix with 1.0 features

Compare the same tasks; keep different tasks separate

Record fine-tuning, data used and setup cost separately for each model



Separate what is known about the future from what is not

You can know next week is an option-expiry week, but not next week's actual volatility



Do not carry model descriptions over to financial performance

Separate general time series results from financial application designs

Model	Caution	Check in a finance course
PatchTST	General time series results cannot stand in for financial superiority	Compare on medium-horizon ETF volume and volatility forecasts
iTransformer	It does not establish causality between variables	Compare alongside a univariate baseline
Foundation models	Zero-shot is not a certificate of financial suitability	Record pretraining period, overlapping data and release date
Chronos-2	Do not attach the value-quantization description of early Chronos	Watch upper-quantile risk signals as well as the median
TimesFM-3	Adding variables can make results worse	Check that no actual future values were input

Keep TOTEM applications and model comparisons separate

Representation-learning tasks and forecasting tasks differ in setup and output

Item	Caution	Correct handling
Codebook	Codes carry no preset economic names	Record tokenizer and task models separately
Anomaly detection	An anomaly signal is not a market-abuse finding	A person checks news, outages, market shocks
Imputation	Imputing from later data is hindsight cleanup	Tell halts and no-trade periods from missing data
Fair comparison	Task-specific training is not a zero-shot condition	Record fine-tuning and setup cost per model
Output type	Do not assume identical quantile or covariate outputs	Separate common and model-specific scores

03

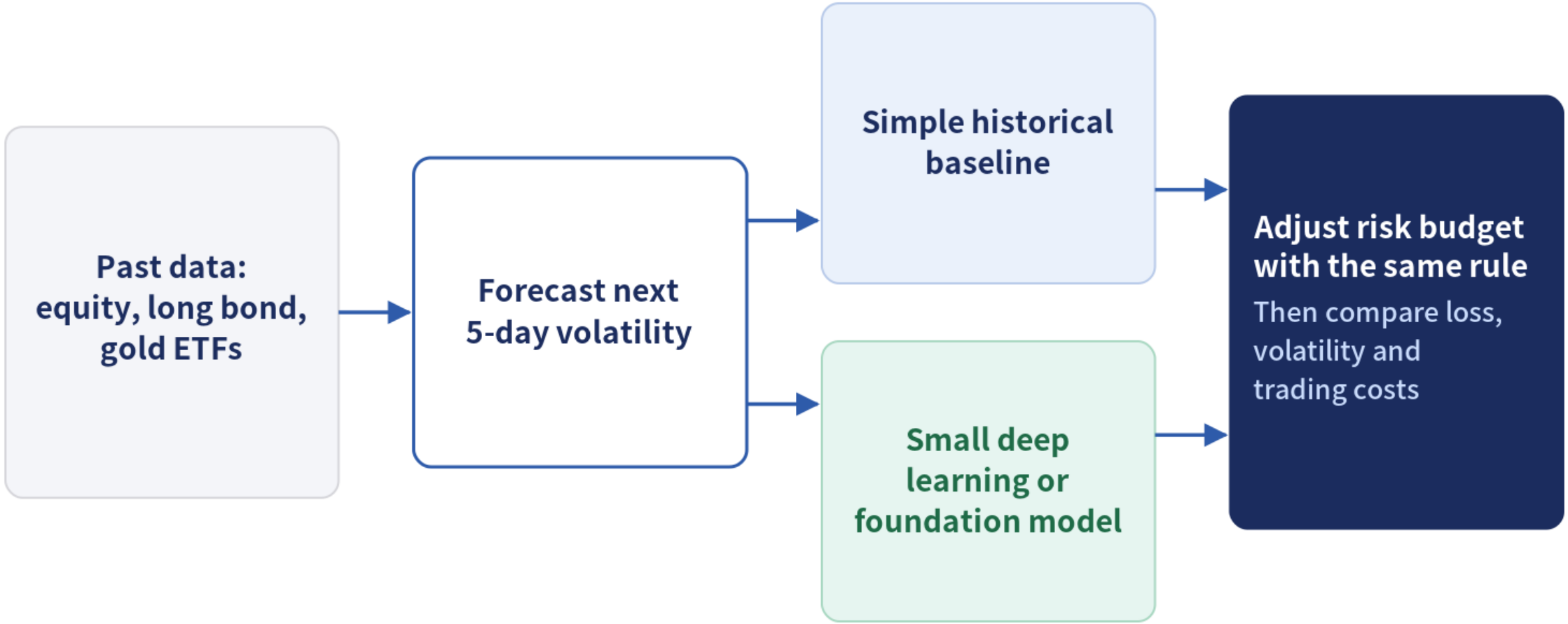
PART 3 · Lab & Evaluation

Financial Forecasting Practice and Time-Ordered Evaluation

Risk budget · order execution · cash flow · time-ordered validation · decision evaluation

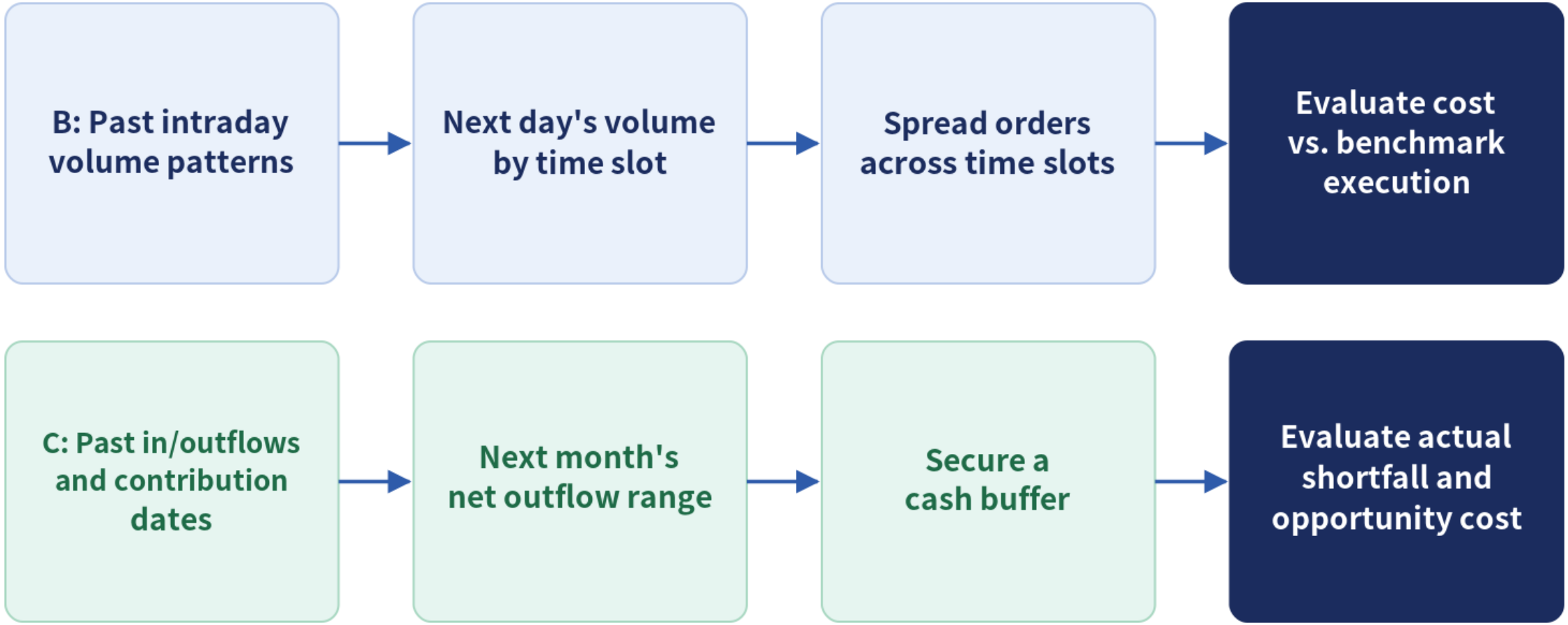
Compare ETF risk budgets with the same rule

Link 5-day volatility forecasts for equity, long bond and gold ETFs to risk-budget adjustment



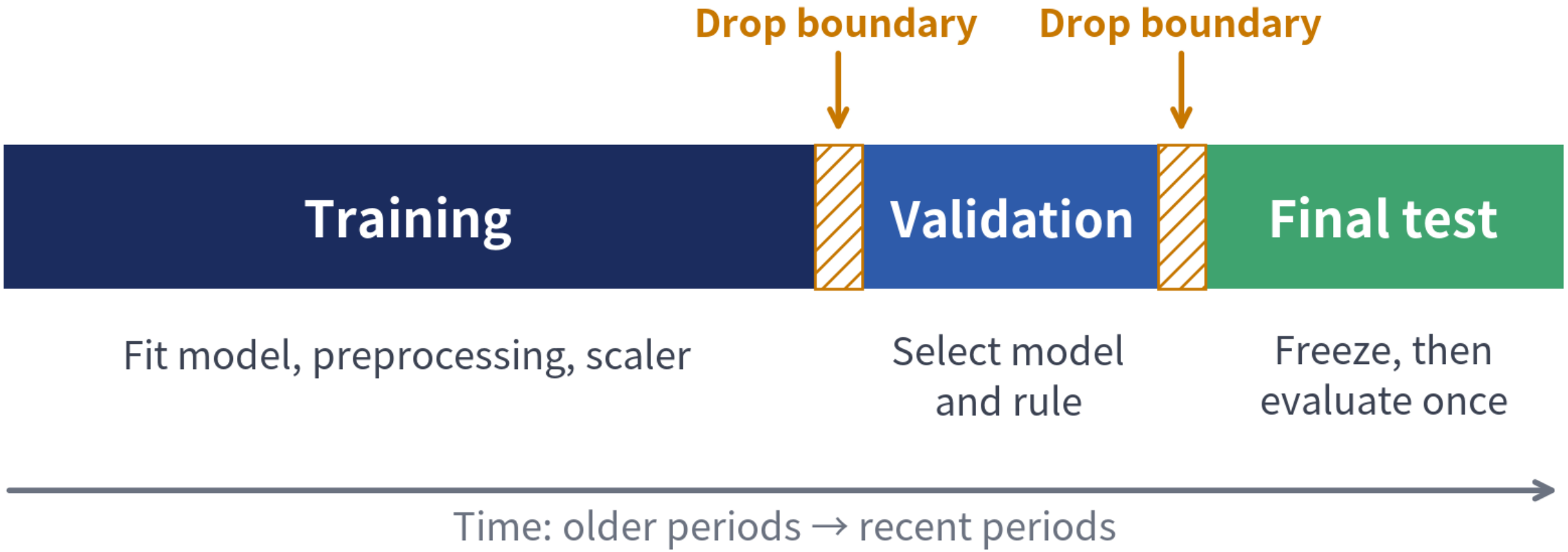
Use volume for execution and net outflows for the cash buffer

Evaluate forecasts by actual costs and shortfalls



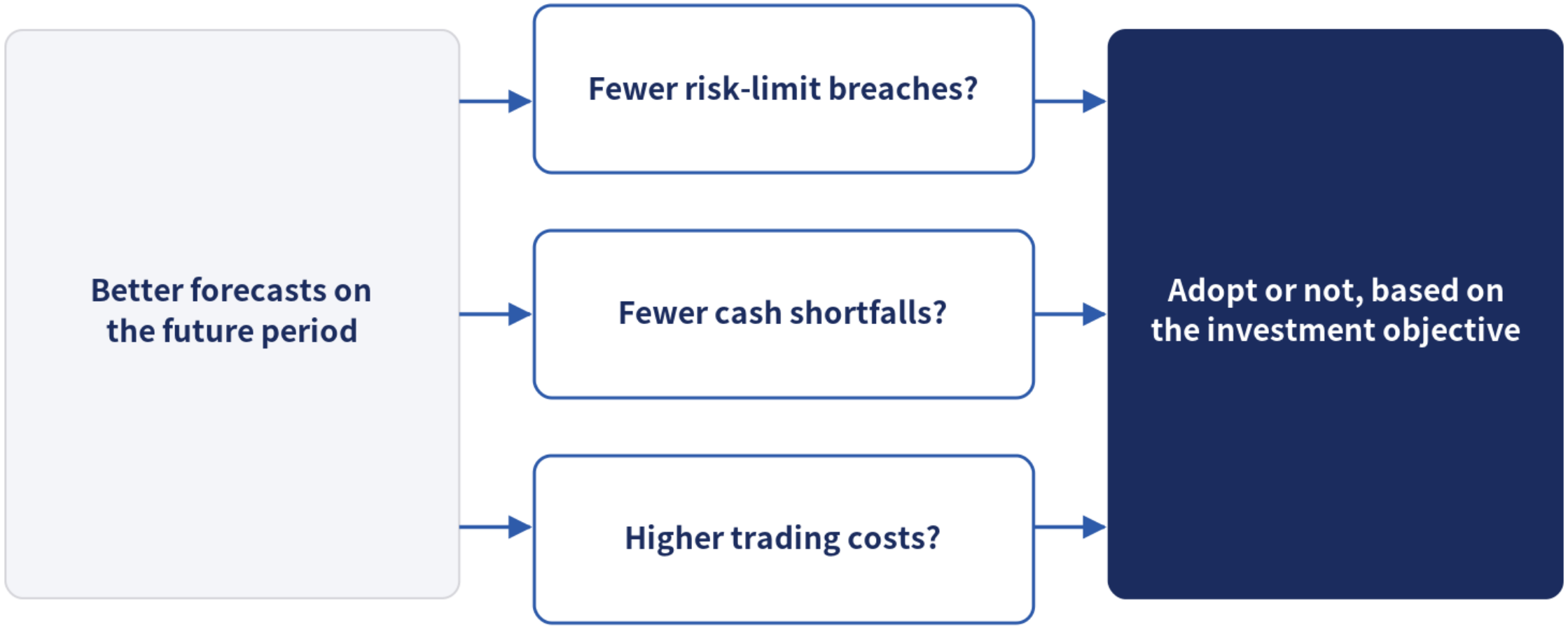
Keep the test period sealed until the end

Split time periods before shuffling any input windows



Ask separately whether a good forecast is a good financial result

Rejudge forecast gains on the future period by risk, cash and cost



Fix rules and costs first in decision evaluation

Check separately whether forecast gains lead to investment gains

Example · procedure	Caution	Correct handling
A Risk budget	Attach a rule that favors each model	Fix in advance: 40% normally, 30% on an upper-band alert
B Order execution	High-volume time slots are low-cost	Account for market impact and spread separately
C Cash flow	An upper-quantile forecast is always safe	Evaluate shortfall reduction and idle-cash cost together
Temporal validation	Shuffle windows first and split later	Split periods first; manage boundary samples and scalers

The report consists of five figures and tables

Do not promise realized excess returns in advance

Input & label periods

Forecast origin, input window and label period in one figure

Architecture figure

Architecture used and input types

Baseline table

Same future period · same label

Coverage figure

Did the stated range contain the actual value?

Decision results

Investment results net of trading costs

Do not equate forecast-score gains with trading-profit gains

Practice: Volatility Forecast Comparison ① Data and Split

Paste into Claude Code — this is one prompt continued in ② on the next slide

[Input] Prompt · Claude Code

```
Write a beginner-level financial time series forecasting exercise.
Use three ETFs (equity, bond, gold) and fix the query end date.
Record the data source, adjusted-price definition, missing values and trading-day
alignment.
If download is not possible, mark prominently that the data are synthetic.

The target is each ETF's realized volatility over the next 5 trading days.
Restrict inputs to the last 60 days of returns, volume and past volatility up to
now.
Explain the label-calculation convention and annualization unit in words.

Split training, validation and test periods in time order first, then build
windows.
Remove boundary samples whose training labels spill into the next period.
Fit scalers and settings only on past periods.
Baselines are a persistence forecast of past realized volatility and a simple
volatility model.
```

What to check

- No future realized values in the inputs?
- Same label compared?
- Boundary samples removed?

Practice: Volatility Forecast Comparison ② Models and Report

Paste right after ①

[Input] Prompt · Claude Code

```
Design a financial forecasting comparison of one small TCN or PatchTST
with Chronos-2, TimesFM-3 and TOTEM.
Run all three models if resources allow; otherwise actually run only one.
For TOTEM, separate the official tokenizer from task-specific models and record
whether it was fine-tuned.
Evaluate TOTEM's anomaly detection and imputation as a separate optional
exercise.
If supported outputs differ across models, separate common forecast scores from
model-specific scores.
Check current official model cards and APIs, and record version, weights and
license.
State whether pretraining data may overlap the test period, and the release date.
Mark models that cannot be run as 'not run' in the results table.

Compare accuracy, interval coverage and compute time on the same future period.
Explain results without formulas, using input-window, architecture and actual-vs-
forecast figures.
Do not equate forecast-score gains with trading-profit gains.
```

Output

- Data record
- Forecast figure
- Model comparison table
- Not-run models marked

Read original papers together with developer announcements

Separate general results, financial experiments and application designs — no model is best at every financial task

No.	Source	Year · status	Scope used in this lecture
S1	PatchTST (arXiv 2211.14730)	ICLR 2023	Patch and channel-independent structure
S2	iTransformer (arXiv 2310.06625)	ICLR 2024	Variable-centric Attention
S3	TimesFM-3 official release (Google Research)	2026-08-31	Recent multivariate foundation model case
S4	Chronos-2 official release (Amazon Science)	2025-10-20	Multivariate · covariates · zero-shot
S10	TOTEM original paper (arXiv 2402.16412)	TMLR 2024 · revised 2025	Discrete tokens · forecasting · anomaly detection · imputation
S11	TOTEM official implementation (GitHub)	Checked 2026-11-04	Separating codebook and task-specific implementations
S12	Datadog Toto official implementation (GitHub)	Checked 2026-11-04	Distinct from TOTEM · features by version

Adopt a forecasting model only when it helps decisions

Next week: instead of forecasting, generate many possible future paths to assess risk

1

Respect the information timing

Split inputs, labels and future covariates at the forecast origin; keep the test period sealed to the end.

2

Record architecture and version

Chronos-2, TimesFM-3 and TOTEM differ in features and setup — record fine-tuning, output type and release date.

3

Compare with simple baselines under the same rule

Judge by accuracy, interval coverage, trading costs and cash shortfalls; if the complex model loses, report that as is.

Week 15 forecasts (conditional) → Week 16 generative Monte Carlo (paths)